

On the Expressive Limits of Formal Truth Predicates

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ABSTRACT

We re-evaluate the axiomatic theories of truth extending Peano Arithmetic. By restricting reflection principles to grounded syntactic formulae, we preserve classical logic while resisting Tarskian semantic hierarchies.

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1. Introduction

Formal theories of truth face classical diagonal limits, as originally formulated by Tarski (1936). A recurrent difficulty concerns whether typed semantic structures are mandatory to avoid paradox (Halbach 2014).

2. Formal preliminaries

Let $\mathcal{L}$ denote the first-order language of arithmetic, and let $\mathcal{L}_T = \mathcal{L} \cup \{T\}$ be the extension obtained by adding a unary truth predicate $T(x)$.

Definition 1 (Arithmetical Coding). For any formula $\varphi \in \mathcal{L}_T$, let $\ulcorner \varphi \urcorner$ denote its canonical Gödel number under a fixed primitive recursive coding scheme.

The standard axiomatization over Peano Arithmetic (PA) requires compositional truth clauses:

$$\text{PA} \models T(\ulcorner \varphi \wedge \psi \urcorner) \iff T(\ulcorner \varphi \urcorner) \wedge T(\ulcorner \psi \urcorner).$$

Theorem 2 (Semantic Triviality). *If a theory $\Sigma \supset \text{PA}$ satisfies unrestricted capture and release:*

$$\Sigma \vdash \varphi \iff T(\ulcorner \varphi \urcorner),$$

then Σ is inconsistent under classical logic.

Proof. By the diagonal lemma, construct a sentence λ such that $\text{PA} \vdash \lambda \iff \neg T(\ulcorner \lambda \urcorner)$. Applying release yields $\Sigma \vdash \lambda \rightarrow \neg \lambda$, from which $\Sigma \vdash \perp$ follows by reductio. $\square$

References

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